

The American Journal of Sports Medicine

<http://ajs.sagepub.com/>

A Randomized Controlled Trial With Mean 16-Year Follow-up Comparing Hamstring and Patellar Tendon Autografts in Anterior Cruciate Ligament Reconstruction

Haukur Björnsson, Kristian Samuelsson, David Sundemo, Neel Desai, Ninni Sernert, Lars Rostgård-Christensen, Jón Karlsson and Jüri Kartus

Am J Sports Med published online May 26, 2016

DOI: 10.1177/0363546516646378

The online version of this article can be found at:

<http://ajs.sagepub.com/content/early/2016/05/26/0363546516646378>

Published by:



<http://www.sagepublications.com>

On behalf of:

American Orthopaedic Society for Sports Medicine



Additional services and information for *The American Journal of Sports Medicine* can be found at:

Published online May 26, 2016 in advance of the print journal.

P<P

Email Alerts: <http://ajs.sagepub.com/cgi/alerts>

Subscriptions: <http://ajs.sagepub.com/subscriptions>

Reprints: <http://www.sagepub.com/journalsReprints.nav>

Permissions: <http://www.sagepub.com/journalsPermissions.nav>

>> [OnlineFirst Version of Record](#) - May 26, 2016

[What is This?](#)

A Randomized Controlled Trial With Mean 16-Year Follow-up Comparing Hamstring and Patellar Tendon Autografts in Anterior Cruciate Ligament Reconstruction

Haukur Björnsson,^{*†‡} MD, Kristian Samuelsson,^{†‡} MD, PhD, David Sundemo,^{†‡} MD, Neel Desai,^{†‡} MD, Ninni Sernert,^{‡§||} RPT, PhD, Lars Rostgård-Christensen,[¶] MD, Jón Karlsson,^{†‡} MD, PhD, and Jüri Kartus,^{‡§||} MD, PhD

Investigation performed at NU-Hospital Group, Trollhättan/Uddevalla, Sweden; Sahlgrenska University Hospital, Mölndal, Sweden; and Capho Artro Clinic, Stockholm, Sweden

Background: There is no consensus in the current literature on which surgical options render the best long-term results after anterior cruciate ligament (ACL) reconstruction in terms of clinical outcomes and the development of radiographic osteoarthritis (OA).

Purpose: To investigate the long-term clinical and radiographic results after ACL reconstruction using either a patellar tendon (PT) autograft or a hamstring tendon (HT) autograft.

Study Design: Randomized controlled trial; Level of evidence, 2.

Methods: This multicenter study was based on 2 previous randomized cohorts consisting of 193 patients who underwent unilateral primary ACL reconstruction using either a PT autograft or an HT autograft. The index surgical procedure was performed between September 1995 and January 2000. Clinical assessments, including knee laxity measurements, functional outcomes, patient-reported outcomes, and bilateral standing radiographs, were performed at the final follow-up examination.

Results: At the long-term follow-up, 147 (76%) patients were examined: 61 in the PT group and 86 in the HT group. The mean ($\pm$ SD) follow-up time was 191.9 $\pm$ 15.1 months for the HT group and 202.6 $\pm$ 10.4 months for the PT group. Knee laxity measurements revealed significantly more patients with a normal pivot-shift test finding in the HT group compared with the PT group (71% vs 51%, respectively; $P = .048$); however, no significant differences were found in terms of the manual Lachman test or the KT-1000 arthrometer manual maximum test. The patients in the PT group had significantly more difficulty knee walking ($P = .049$). There were no significant differences between the study groups in terms of patient-reported outcomes or range of motion in the reconstructed knee. In both groups, significantly more signs of radiographic OA were found in the reconstructed knee than in the contralateral healthy knee. However, there were no significant differences between the groups in terms of radiographic OA.

Conclusion: Only minor and mostly insignificant differences were found between the PT and HT autograft groups in this long-term randomized controlled trial. In both groups, significantly more signs of radiographic OA were found in the reconstructed knee than in the contralateral knee.

Keywords: anterior cruciate ligament; reconstruction; surgery; patellar tendon; semitendinosus; gracilis; long-term follow-up; osteoarthritis; randomized trial; radiographs

Graft selection for anterior cruciate ligament (ACL) reconstruction has been an intense research topic and debate subject for several decades. Hamstring tendon (HT) and patellar tendon (PT) autografts are currently the most utilized grafts in both research and clinical practice, although allografts and quadriceps tendon autografts are still the preferred choice.³³ Both HT and PT autografts are easy to access in the surgical setting when performing ACL reconstruction. The PT

autograft has its advantages because of the bone blocks at both ends of the graft, which facilitate ingrowth of the graft. However, the quadruple HT autograft is both a stiffer and stronger graft choice, with easy-to-manage harvest site morbidity.³⁴ Even though the quadruple HT autograft is stronger, concerns have been raised about the risk of increased knee laxity over time.

Looking beyond differences in harvest site symptoms, graft-bone healing, and strength or stiffness issues relating to the autografts, there are 3 outcomes that are crucial for the patient: graft rupture, return to play, and the development of osteoarthritis (OA). Previous studies have reported that there are no differences in terms of rupture frequency

between the 2 autografts.^{2,6,7,11,14,15,27,37} However, recent register studies have contested these results and indicated a higher rupture frequency in patients undergoing surgery using HT autografts.^{10,19,20,26,28} Even though previously reported data are overwhelming, it could be interesting to ask why recent large register studies have revealed contradictory results. Register studies are usually robust, but noncalculated confounders might be one cause of these recent findings.

Patients who have sustained an ACL injury run the risk of developing posttraumatic OA, with the first signs of radiographic joint space narrowing at the age of as young as 40 years.³¹ The incidence of radiographic OA 10 to 20 years after an ACL injury has been estimated at approximately 50%,¹⁸ with a higher incidence in patients with combined injuries compared with isolated ACL ruptures.^{22,23} Previous studies including systematic reviews have reported somewhat contradictory results in terms of the autograft that yields the highest risk of developing posttraumatic OA.^{15,34,36} To date, however, there are no reported randomized controlled trials (RCTs) with more than 15 years' follow-up comparing the 2 autografts in terms of the development of OA.

The purpose of this long-term RCT was to compare the HT autograft and PT autograft in terms of patient-reported outcome measures, functional outcomes, graft failures, clinical evaluations including knee laxity measurements, and radiographic evaluations. The hypothesis was that both graft types would render a similar outcome in the long term.

METHODS

Patients

This long-term study consisted of 2 previous randomized trials^{6,14} with patients who had sustained a unilateral ACL rupture and underwent arthroscopic ACL reconstruction using either ipsilateral HT or PT autografts (Figure 1). Patients were eligible in the original cohorts if they had an isolated ACL rupture with or without additional meniscal or minor chondral lesions (Outerbridge grades 1 and 2). Patients were excluded if they had multiligament injuries, had undergone previous ACL reconstruction, or had chondral lesions that mandated surgical intervention other than debridement. Patients were randomized preoperatively using nontransparent, white sealed envelopes to either ACL reconstruction with ipsilateral triple semitendinosus tendon (ST) autografts, ipsilateral quadruple ST autografts, ipsilateral quadruple ST and gracilis tendon (ST/GT) autografts, or ipsilateral PT autografts. The HT group consisted of 3 different graft selections because of

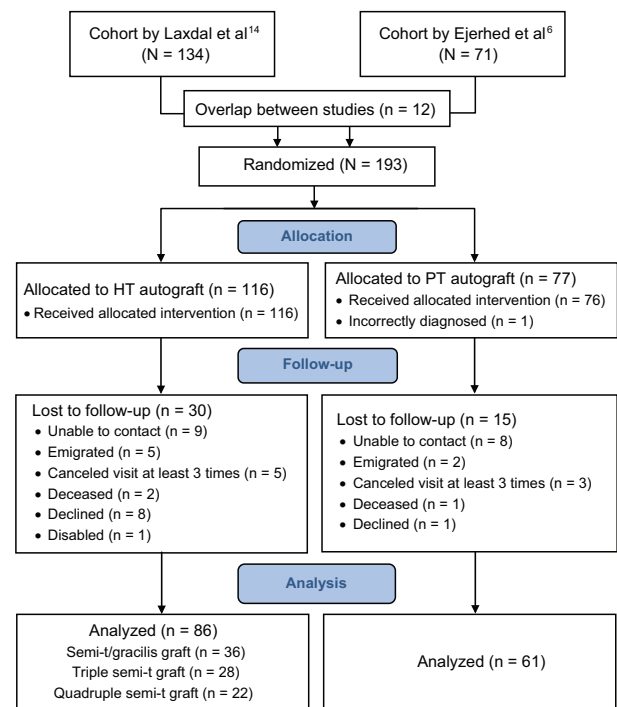


Figure 1. Flowchart of included patients. Semi-t, semitendinosus.

the consolidation of 2 previous RCTs. Reconstruction was performed between September 1995 and January 2000 at 3 different centers by 1 of 6 experienced surgeons with long, well-documented experience of ACL reconstruction. In this long-term study, all the patients who had undergone reconstruction using either 1 or 2 HTs were analyzed in the same group (Figure 1). The Human Ethics Committee of the Medical Faculty at Gothenburg University and Stockholm University approved the study, and all patients gave their informed and written consent.

Surgical Technique

All patients received preoperative antibiotic prophylaxis, and surgery was performed with the patient in the supine position under general or spinal anesthesia. Standard arthroscopic portals were used in all patients. Associated injuries were treated with debridement, resection, or repair.

PT Autograft. The central third of the PT was harvested either through an open approach with a 60 mm– to 80 mm–long vertical incision (n = 31) or through 2 approximately 25 mm–long vertical incisions (n = 30) with subcutaneous graft tunneling under the paratenon to protect the infrapatellar nerve branches of the saphenous nerve, leaving the

*Address correspondence to Haukur Björnsson, MD, Department of Orthopaedics, Institute of Clinical Sciences, Sahlgrenska Academy, University of Gothenburg, Bruna Stråket 11, Vän 5, SU/Sahlgrenska, Göteborg, 413 45, Sweden (email: haukurbj@hotmail.com).

[†]Department of Orthopaedics, Sahlgrenska University Hospital, Mölndal, Sweden.

[‡]Department of Orthopaedics, Institute of Clinical Sciences, Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden.

[§]Department of Research and Development, NU-Hospital Group, Trollhättan/Uddevalla, Sweden.

^{||}Department of Orthopaedics, NU-Hospital Group, Trollhättan/Uddevalla, Sweden.

[¶]Department of Radiology, Skaraborg Hospital, Lidköping, Sweden.

The authors declared that they have no conflicts of interest in the authorship and publication of this contribution.

major part of the paratenon intact. The tendon defect was left open, and defects in the patella and the proximal tibia were not bone grafted. The proximal bone block was sized at 9 mm, and the distal bone block was sized at 10 mm. The femoral bone tunnels were created through transtibial drilling, aiming at the 10:00 to 10:30 clock position. The tibial tunnel was drilled with an elbow aimer in the native ACL footprint. Seven- and 9-mm interference screws (Acufex Microsurgical) were used on the femoral and tibial sides, respectively. The knee was hyperextended, and firm traction was applied to the autograft before fixation of the tibial interference screw.

Triple and Quadruple ST Autografts. The graft was harvested through a 3-cm oblique incision over the pes anserinus. The sartorius fascia was incised parallel to the fibers of the fascia, just above the thicker and more distally inserted ST. After the vincula were divided, the tendon was harvested using a semiblunt, semicircular open tendon stripper (Acufex Microsurgical). The tendon was constructed to create a 3- or 4-stranded graft measuring a minimum of 7 cm in length. Two No. 5 nonresorbable Ti-Cron sutures (Sherwood Medical) were used as the lead sutures at the distal and proximal ends. Resorbable No. 1 Vicryl sutures (Ethicon Inc) were used for the modified baseball stitches at the distal and proximal ends of the ST graft. The femoral tunnel was drilled transtibially or through the medial portal. Bone tunnels were drilled using specially designed Stepped Routers (Smith & Nephew plc), aiming at the 10:00 to 10:30 clock position. The tibial tunnel was drilled with an elbow aimer in the native ACL footprint. A 7-mm, soft-threaded, round-headed RCI interference screw (Smith & Nephew plc) was used on both the femoral and tibial sides. The knee was hyperextended, and firm traction was applied to the autograft before fixation of the tibial interference screw.

Quadruple ST/GT Autograft. The surgical technique for the quadruple ST/GT autograft group was identical to that for the triple and quadruple ST autograft groups, except that both the GT and ST were harvested and constructed to create a 4-stranded autograft.

Rehabilitation

All patients underwent a similar rehabilitation program with immediate full weightbearing, without a brace, and with full range of motion (ROM) utilizing the weight of the leg. External loads were not permitted from 30° to hyperextension for the first 6 weeks. Closed chain exercises were started immediately postoperatively. Running was permitted at 3 months and contact sports at 6 months at the earliest, if the patient had regained full functional stability in terms of strength, coordination, and balance as compared with the contralateral leg.

Clinical Assessments and Follow-up

Independent physical therapists performed the preoperative assessments, and 1 research assistant with thorough knee evaluation knowledge performed all the long-term follow-up assessments. None of them had been involved in the surgery or rehabilitation of the patients. A KT-1000 arthrometer (MEDmetric Corp) was used for the

assessment of anteroposterior knee laxity. The Lysholm score, Tegner activity score, and International Knee Documentation Committee (IKDC) evaluation system were utilized to evaluate functional outcomes. ROM was measured to the nearest 5° with a goniometer. A disturbance in sensitivity in the anterior region of the knee was measured (in cm²) by palpation. The kneeling discomfort classification was based on a 4-grade knee-walking test in which the patients walked on the floor on their knees without any protective clothing. Moreover, the 1-legged hop test was used to evaluate functional performance, and a quotient (%) between the injured leg and uninjured leg was calculated.

Radiographic Assessments

At the long-term follow-up, standard weightbearing radiographic assessments were performed on both knees. The radiographs were assessed and classified according to the Fairbank,⁸ Kellgren-Lawrence,¹³ and Ahlback¹ scores by an independent senior radiologist who specializes in orthopaedic radiographic evaluations. The intrarater reproducibility analyses for the radiologist in the study revealed kappa values between .55 and 1.00 when classifying and reclassifying the results of the Ahlback and Fairbank rating systems for 20 randomly selected patients.¹⁷ For the Fairbank system, the cumulative number of positive findings, from 0 to 6, was calculated for each patient.¹⁷ OA changes in the patellofemoral joint were classified as none (0), minor (1), moderate (2), or severe (3). Correspondingly, patellofemoral osteophytes were classified as none (0), small (1), moderate (2), or large (3).

Statistical Analyses

The triple and quadruple HT autograft groups revealed no differences at the short-term follow-up.¹⁴ So, to increase the sizes of the study groups, the patients were classified as reconstructed either with a PT autograft or with an HT autograft at the long-term follow-up. Data are presented as medians and ranges. In terms of both continuous and noncontinuous variables, the Mann-Whitney *U* test was used. The Wilcoxon signed-rank test was used for comparisons of the preoperative and postoperative data within the groups. The chi-square test was used to compare the dichotomous variables between the study groups. Alpha was set at .05. At the time when the studies were initiated (before 1995), power analyses were not performed.

RESULTS

A total of 193 patients (62 women and 131 men) were included in the study, of which 147 (76%) patients were examined at the long-term follow-up: 61 patients in the PT group (19 female and 42 male) and 86 patients in the HT group (33 female and 53 male) (Table 1). The mean ($\pm$ SD) follow-up time was 191.9 $\pm$ 15.1 months for the HT group and 202.6 $\pm$ 10.4 months for the PT group (*P* < .001). Both groups were similar in terms of sex, age at surgery, time between injury and surgery,

TABLE 1
Descriptive Data^a

	PT Group (n = 61)	HT Group (n = 86)	P Value
Sex, female/male, n	19/42	33/53	.37
Age at surgery, y			.54
Mean $\pm$ SD	28.2 $\pm$ 9.1	26.8 $\pm$ 7.6	
Median (range)	26 (14-52)	26 (15-59)	
Time between injury and surgery, mo			.48
Mean $\pm$ SD	29.9 $\pm$ 46.9	38.1 $\pm$ 60.4	
Median (range)	11.5 (2-252)	14.0 (2-360)	
Age at follow-up, y			.17
Mean $\pm$ SD	44.7 $\pm$ 9.1	42.3 $\pm$ 7.8	
Median (range)	43 (31-68)	42 (29-74)	
Associated injuries (meniscal/chondral lesions)			.59
Yes/no, n (%)	39/22 (63.9/36.1)	58/27 (67.4/31.4)	
Meniscal repair, n	2	2	
Missing, n	—	1	
Additional surgery during follow-up, yes/no, n (%)	32/29 (52.5/47.5)	42/44 (48.8/51.2)	.67
Cause of additional surgery			
ACL graft rupture	4 (6.6)	7 (8.1)	
Chondral lesion	2 (3.3)	3 (3.5)	
Extension deficit	1 (1.6)	—	
Meniscal problem	14 (23.0)	21 (24.4)	
Other	7 (11.5)	8 (9.3)	
PCL injury	1 (1.6)	—	
Screw problem	2 (3.3)	2 (2.3)	
Septic knee	1 (1.6)	1 (1.2)	
Surgery in contralateral knee, yes/no, n (%)	11/50 (18.0/82.0)	12/74 (14.0/86.0)	.50
Cause of surgery			
ACL injury	6 (9.8)	6 (7.0)	
Chondral lesion	—	1 (1.2)	
Meniscal problem	4 (6.6)	4 (4.7)	
Other	1 (1.6)	1 (1.2)	
Follow-up, mo			<.001
Mean $\pm$ SD	202.6 $\pm$ 10.4	191.9 $\pm$ 15.1	
Median (range)	203 (164-224)	197 (159-225)	

^aBolded *P* values indicate statistically significant difference between groups. ACL, anterior cruciate ligament; HT, hamstring tendon; PCL, posterior cruciate ligament; PT, patellar tendon.

and associated injuries (Table 1). No differences were found in terms of cause of injury; the most common cause of injury was contact sport, with 70.5% in the PT group and 66.3% in the HT group (Table 2).

Graft Rupture

No differences were found between the 2 groups in terms of additional surgery during follow-up and surgery on the contralateral knee (Table 1). The most frequent cause of additional surgery on the index knee was a meniscal problem, while it was an ACL injury in the contralateral knee. Four patients (6.6%) in the PT group and 7 patients (8.1%) in the HT group had an ACL graft rupture. Six patients in both groups (PT: 9.8%, HT: 7.0%) sustained an ACL injury to the contralateral knee (Table 1).

Knee Laxity Measurements

Patients in the HT group had significantly higher preoperative manual Lachman values ($P = .021$); this was also

TABLE 2
Causes of Injury^a

	PT Group (n = 61)	HT Group (n = 86)
Activities of daily living	2 (3.3)	2 (2.3)
Contact sport	43 (70.5)	57 (66.3)
Noncontact sport	8 (13.1)	12 (14.0)
Other	6 (9.8)	7 (8.1)
Work	1 (1.6)	0 (0)
Missing values	1 (1.6)	8 (9.3)

^aData are reported as n (%). There were no statistically significant differences between groups ($P = .8$). HT, hamstring tendon; PT, patellar tendon.

found when patients with an ACL injury to the index knee or the contralateral knee were excluded from the analysis ($P = .042$) (Table 3 and Appendix Table A1, available online at <http://ajsm.sagepub.com/supplemental>). Both groups had significantly less knee laxity postoperatively in terms of the Lachman test (PT: $P < .0001$, HT: $P < .0001$) and the pivot-shift test (PT: $P = .001$, HT: $P < .0001$).

TABLE 3
Knee Laxity Measurements^a

	PT Group (n = 61)	HT Group (n = 86)	P Value
Lachman grade			
Preoperative			.021
0	1 (1.6) ^b	—	
1	10 (16.4)	19 (22.1) ^b	
2	26 (42.6)	46 (53.5)	
3	22 (36.1)	12 (14.0)	
Missing	2 (3.3)	9 (10.5)	
Follow-up			.11
0	30 (49.2)	34 (39.5)	
1	24 (39.3)	32 (37.2)	
2	6 (9.8)	17 (19.8)	
3	1 (1.6)	3 (3.5)	
Pivot-shift grade			
Preoperative			.54
0	7 (11.5) ^b	9 (10.5) ^b	
1	44 (72.1)	50 (58.1)	
2	6 (9.8)	15 (17.4)	
3	2 (3.3)	—	
Missing	2 (3.3)	12 (14.0)	
Follow-up			.21
0	34 (55.7)	58 (67.4)	
1	14 (23.0)	15 (17.4)	
2	12 (19.7)	12 (14.0)	
3	—	1 (1.2)	
Missing	1 (1.6)	—	
KT-1000 arthrometer side-to-side difference, ^c mm			
Follow-up: 134 N			.70
<-3	4 (6.6)	4 (4.7)	
-3 to 0	25 (41.0)	30 (34.9)	
0 to 3	19 (31.1)	27 (31.4)	
>3	13 (21.3)	25 (29.1)	
Follow-up: MMT			.35
<-3	4 (6.6)	3 (3.5)	
-3 to 0	23 (37.7)	23 (26.7)	
0 to 3	20 (32.8)	33 (38.4)	
>3	14 (23.0)	27 (31.4)	
Preoperative: 89 N			.96
Mean ± SD	4.1 ± 4.0	4.3 ± 4.0	
Median (range)	4.0 (-4.0 to 17.0)	3.5 (-2.0 to 24.0)	
Missing, n	3	9	
Follow-up: 134 N			.32
Mean ± SD	1.0 ± 2.7	1.5 ± 3.0	
Median (range)	1.0 (-5.0 to 6.0)	1.5 (-6.0 to 8.0)	
Follow-up: MMT			.14
Mean ± SD	1.0 ± 2.8	1.8 ± 3.0	
Median (range)	1.0 (-5.0 to 7.0)	2.0 (-7.0 to 8.0)	

^aData are reported as n (%) unless otherwise indicated. Bolded *P* values indicate statistically significant difference between groups. HT, hamstring tendon; MMT, manual maximum test; PT, patellar tendon.

^bSignificant difference between the preoperative and follow-up values in both groups.

^cPreoperative values were measured at 89 N, while 134 N and MMT were used for postoperative measurements.

.0001). The pivot-shift test revealed significantly more patients with a normal pivot shift in the HT group when reinjured patients and patients with a contralateral ACL injury were excluded (*P* = .048) (Appendix Table A3, available online). Otherwise, no differences were found in terms of the Lachman test, the pivot-shift test, or the KT-1000 arthrometer measurements.

Clinical Assessments

Both patient groups had reduced ROM in both flexion and extension at follow-up compared with the preoperative values. However, there were no differences in ROM between the groups in the injured knee at follow-up (Appendix Tables A2 and A3, available online).

TABLE 4
Clinical Assessments^a

	PT Group (n = 61)	HT Group (n = 86)	P Value
Knee-walking test			
Preoperative			.61
OK	27 (44.3)	38 (44.2)	
Not pleasant	24 (39.3)	29 (33.7)	
Difficult	7 (11.5)	7 (8.1)	
Impossible	2 (3.3)	3 (3.5)	
Missing	1 (1.6)	9 (10.5)	
Follow-up			.049
OK	30 (49.2)	54 (62.8)	
Not pleasant	15 (24.6)	21 (24.4)	
Difficult	8 (13.1)	5 (7.0)	
Impossible	8 (13.1)	6 (5.8)	
One-legged hop test			
Preoperative			.13
Mean ± SD	79.0 ± 23.5	72.3 ± 29.5 ^b	
Median (range)	88 (0-111)	82 (0-113)	
Missing, n	2	9	
Follow-up			.13
Mean ± SD	82.2 ± 33.3	88.9 ± 25.2	
Median (range)	89 (0-184)	93 (0-160)	
Disturbance in knee sensitivity, cm ²			
Preoperative			.12
Mean ± SD	8.0 ± 27.3 ^c	3.8 ± 19.1 ^c	
Median (range)	0 (0-168)	0 (0-143)	
Missing, n	1	12	
Follow-up			.48
Mean ± SD	59.6 ± 74.2	69.1 ± 83.3	
Median (range)	28 (0-288)	31 (0-340)	

^aData are reported as n (%) unless otherwise indicated. Bolded *P* values indicate statistically significant difference between groups. HT, hamstring tendon; PT, patellar tendon.

^bSignificant difference between preoperative and follow-up assessments in the HT group.

^cSignificant difference between preoperative and follow-up assessments in both groups.

Patients in the HT group increased their performance in the 1-legged hop test (PT: *P* = .382, HT: *P* < .0001) at follow-up compared with the preoperative values. Both groups had an increased disturbance in knee sensitivity compared with the preoperative values (PT and HT: *P* < .0001). Patients in the HT group had better knee-walking ability (*P* = .049). This difference was also seen when patients with a new ACL injury on the same or contralateral side were excluded (*P* = .03) (Table 4 and Appendix Table A4, available online).

Subjective Evaluations

Postoperatively, both patient groups had a significant increase in the Lysholm score (PT: *P* = .002, HT: *P* < .0001) compared with preoperative values. Postoperatively, there was a significant decrease in terms of the Tegner activity score compared with the preinjury scores (PT and HT: *P* < .0001). The same finding was evident when patients with a graft rupture and a contralateral ACL rupture were excluded. There were no significant differences between the 2 groups in terms of the subjective IKDC score (Table 5 and Appendix Table A5, available online).

Radiographic Evaluations of OA

Both groups revealed significantly more postoperative radiographic changes in the injured knee compared with the contralateral knee (*P* < .001) (Table 6 and Appendix Table A6, available online). These results were consistent, regardless of which classification system was used and whether patients with a new ACL injury on the index side or contralateral side were excluded. No significant differences were found between the study groups.

DISCUSSION

The most important finding in the present long-term study was that both groups revealed significantly more radiographic OA in the injured knee compared with the healthy contralateral knee. Otherwise, few differences were found when comparing the results after ACL reconstruction using either HT or PT autografts.

A graft rupture is of major importance to the patient in both the short and long term, and for this reason, an effort to minimize this risk would be of great value. Revision surgery and graft ruptures have recently attracted more

TABLE 5
Subjective Evaluations^a

	PT Group (n = 61)	HT Group (n = 86)	P Value
Lysholm knee score			
Preoperative			.91
Mean $\pm$ SD	70.2 $\pm$ 14.4 ^b	70.3 $\pm$ 14.3 ^b	
Median (range)	71 (14-95)	73 (21-99)	
Missing, n	1	9	
Follow-up			.77
Mean $\pm$ SD	79.4 $\pm$ 16.9	80.7 $\pm$ 15.3	
Median (range)	83 (32-100)	85 (8-100)	
Tegner activity score			
Preinjury			.50
Mean $\pm$ SD	8.0 $\pm$ 1.4 ^c	7.9 $\pm$ 1.4 ^c	
Median (range)	9 (3-10)	9 (4-10)	
Missing, n	1	10	
Preoperative			.86
Mean $\pm$ SD	4.0 $\pm$ 1.6	3.9 $\pm$ 1.3	
Median (range)	4 (1-9)	4 (2-9)	
Missing, n	1	9	
Follow-up			.84
Mean $\pm$ SD	4.1 $\pm$ 1.7	4.0 $\pm$ 1.7	
Median (range)	4 (1-10)	4 (0-8)	
IKDC subjective score (total) at follow-up			.05
Mean $\pm$ SD	67.3 $\pm$ 20.8	74.0 $\pm$ 18.8	
Median (range)	71.0 (12.6-98.8)	78.5 (11.5-100)	

^aHT, hamstring tendon; IKDC, International Knee Documentation Committee; PT, patellar tendon.^bSignificant difference between preoperative and follow-up assessments in both groups.^cSignificant difference between preinjury and follow-up assessments in both groups.

attention because of recent register studies that have shown a significantly increased risk of revision surgery in patients who have received an HT autograft.^{10,20,26,28} Data from the present study did not reveal any increase in the risk in terms of graft ruptures in patients with an HT autograft in the long term. However, the reason for the proposed increased risk of revision surgery in patients who have undergone HT autograft reconstruction in the above-mentioned studies might not be graft related. A confounder might be involved, which is yet to be identified.

In the present study, the second most common reason for revision surgery, after a meniscal problem, was a second ACL injury. Not only do the data in the present study contradict the 5-year data from the above-mentioned register studies, but they also suggest that both autograft groups run twice the risk of revision surgery in the long term. The reasons for this might be the 3 times longer follow-up period in the present RCT and possibly the high loss to follow-up in the registers. Moreover, the results from the present study are reproduced in the majority of published studies, including long-term studies, systematic reviews, meta-analyses, and a Cochrane review.[#]

In the present study, patients with contralateral ACL injuries were approximately as common as patients who had sustained a graft rupture. Two other long-term studies with 15-year follow-ups have assessed this issue.^{4,15} Bourke et al⁴ reported results similar to those in the present study; however, the study by Leys et al¹⁵ found that

patients who had undergone PT autograft reconstruction sustained more than twice as many contralateral injuries at the final follow-up: 26% versus 12% for HT autografts. The Leys et al¹⁵ study was not an RCT, and a selection bias may therefore be present.

In the present study, few differences were found between the 2 groups in terms of knee laxity measurements. This is in line with previous studies on the subject.³⁴ Both groups had significantly less knee laxity postoperatively, indicating that reconstruction was successful in restoring macroscopic knee kinematics and laxity. Furthermore, the clinical assessments revealed very few significant differences between the 2 study groups. Several previous studies, including short-term reports on the same cohorts, only found differences related to harvest site morbidity, with increased morbidity in patients undergoing PT autograft reconstruction.^{6,14} In the present study, the differences between the 2 groups in terms of harvest site morbidity were small but statistically significant. The reasons for the small differences might be that harvest site morbidity from PT autograft harvesting diminishes with time and that patients in both groups start to develop degenerative changes, thus minimizing the difference between groups. Moreover, some patients had their PT autograft harvest through a potential nerve-sparing 2-incision technique. The development of degenerative changes might also be the reason for both groups having less ROM at the follow-up compared with the preoperative values.

Both groups had a significant postoperative increment in the Lysholm score and a decrease in the Tegner activity

[#]References 3, 9, 12, 15, 16, 21, 27, 29, 32, 35.

TABLE 6
Radiographic Evaluations at Follow-up^a

	PT Group (n = 61)		HT Group (n = 86)		P Value
	Injured	Contralateral	Injured	Contralateral	
Ahlbäck score					
Medial					Injured: .65 Contralateral: .40
0	40 (65.6) ^b	56 (91.8)	61 (70.9) ^b	82 (95.3)	
1	15 (24.6)	3 (4.9)	15 (17.4)	1 (1.2)	
2	3 (4.9)	—	6 (7.0)	—	
3	1 (1.6)	—	1 (1.2)	—	
4	—	—	1 (1.2)	—	
5	—	—	—	1 (1.2)	
Missing	2 (3.3)	2 (3.3)	2 (2.3)	2 (2.3)	
Lateral					Injured: .25 Contralateral: .67
0	40 (65.6) ^b	56 (91.8)	64 (74.4) ^b	81 (94.2)	
1	15 (24.6)	3 (4.9)	17 (19.8)	2 (2.3)	
2	3 (4.9)	—	2 (2.3)	—	
3	1 (1.6)	—	1 (1.2)	1 (1.2)	
Missing	2 (3.3)	2 (3.3)	2 (2.3)	2 (2.3)	
Cumulative Fairbank score changes ^c					Injured: .53 Contralateral: .39
Mean ± SD	2.4 ± 1.7 ^b	0.5 ± 0.95	2.1 ± 1.8 ^b	0.4 ± 0.96	
Median (range)	2 (0-6)	0 (0-3)	2 (0-6)	0 (0-6)	
Missing, n	2	2	2	2	
Patellofemoral arthrosis ^d					Injured: .14 Contralateral: .37
0	51 (83.6)	54 (88.5)	78 (90.7)	80 (93.0)	
1	7 (11.5)	1 (1.6)	3 (3.5)	1 (1.2)	
2	1 (1.6)	3 (4.9)	1 (1.2)	2 (2.3)	
3	—	1 (1.6)	1 (1.2)	1 (1.2)	
Missing	2 (3.3)	2 (3.3)	3 (3.5)	2 (2.3)	
Patellofemoral osteophytes ^e					Injured: .67 Contralateral: .41
0	30 (49.2) ^b	48 (78.7)	45 (52.3) ^b	73 (84.9)	
1	23 (37.7)	11 (18.0)	31 (36.0)	9 (10.5)	
2	6 (9.8)	—	6 (7.0)	1 (1.2)	
3	—	—	1 (1.2)	1 (1.2)	
Missing	2 (3.3)	2 (3.3)	3 (3.5)	2 (2.3)	
Kellgren-Lawrence score					Injured: .53 Contralateral: .39
0	17 (27.9) ^b	47 (77.0)	26 (30.2) ^b	70 (81.4)	
1	14 (23.0)	5 (8.2)	25 (29.1)	11 (12.8)	
2	22 (36.1)	7 (11.5)	23 (26.7)	2 (2.3)	
3	6 (9.8)	—	8 (9.3)	—	
4	2 (3.3)	—	4 (4.7)	1 (1.2)	
Missing	—	2 (3.3)	—	2 (2.3)	

^aData are reported as n (%) unless otherwise indicated. HT, hamstring tendon; PT, patellar tendon.

^bSignificant difference between injured and contralateral sides in both groups ($P < .001$).

^cThe cumulative number of positive findings, from 0 to 6, was calculated for each patient.

^dClassified as none (0), minor (1), moderate (2), or severe (3).

^eClassified as none (0), small (1), moderate (2), or large (3).

score; however, no significant differences were found between the groups. The Tegner activity score had decreased to half from the preinjury values. The reason for this is certainly changes in lifestyle and preferences when aging, in addition to increased degenerative changes in the knee. The Tegner activity score was lower than that found in age-related

patients with healthy knees,⁵ indicating that an ACL-injured and reconstructed knee limits patients more than normal aging.

At the long-term follow-up, both groups revealed significantly more radiographic OA in the injured knee compared with the contralateral knee. OA is the leading

cause of knee-related pain and disability in middle-aged people.^{25,38} For this reason, preventing the development of posttraumatic OA is of major importance. Previous studies have been inconclusive about the benefits of any kind of autograft in terms of minimizing the risk of future OA. Some studies have pointed toward an increase in the development of OA in patients reconstructed with PT autografts.^{27,30} This finding was not reproduced in the present study. It appears that when everything else is similar, the choice of autograft does not affect the outcomes of long-term knee health. There is, of course, an issue with the lack of correlation between radiographic signs and clinical symptoms²⁴; the symptoms and radiographs should therefore be assessed separately.

On the basis of the results of the present study, we propose that the graft choice for ACL reconstruction should be based on the preference of the patient and the surgeon rather than possible minor differences in the short- and long-term outcomes.

The limitations of the present study involve the follow-up rate, which is just below 80%, and the fact that it consists of a merger of 2 previous randomized studies. It is also an obvious limitation that the HT autografts in the previous studies were either a quadruple ST/GT or a triple or quadruple ST. Moreover, the PT autograft harvest technique in the previous studies was not consistent. Another limitation is that no sample size calculations were performed before the initiation of the studies. The fact that the KT-1000 arthrometer anterior side-to-side difference was measured at only 89 N preoperatively, as opposed to 134 N and manual maximum at follow-up, is another weakness. That no interreliability and intrareliability analyses were performed for the pivot-shift test and that return to sports was not considered are, of course, limitations as well. Furthermore, the use of both the transtibial and medial portal approaches during femoral tunnel drilling must also be regarded as a potential limitation. However, at the time when the patients underwent reconstruction, the anatomic principles were not known, and both techniques aimed to place the femoral tunnels in the same position. Finally, it is also important to consider that the nonsignificant differences that were found could still be caused by a type II error.

CONCLUSION

Only minor and mostly insignificant differences were found between the PT and HT autograft groups in this long-term RCT. In both groups, significantly more signs of radiographic OA were found in the reconstructed knee than in the contralateral healthy knee.

REFERENCES

1. Ahlbäck S. Osteoarthritis of the knee: a radiographic investigation. *Acta Radiol Diagn*. 1968;Suppl 277:7-72.
2. Bärenius B, Nordlander M, Ponzer S, Tidermark J, Eriksson K. Quality of life and clinical outcome after anterior cruciate ligament reconstruction using patellar tendon graft or quadrupled semitendinosus

- graft: an 8-year follow-up of a randomized controlled trial. *Am J Sports Med*. 2010;38(8):1533-1541.
3. Biau DJ, Tournoux C, Katsahian S, Schranz PJ, Nizard RS. Bone-patellar tendon-bone autografts versus hamstring autografts for reconstruction of anterior cruciate ligament: meta-analysis. *BMJ*. 2006;332(7548):995-1001.
4. Bourke HE, Salmon LJ, Waller A, Patterson V, Pinczewski LA. Survival of the anterior cruciate ligament graft and the contralateral ACL at a minimum of 15 years. *Am J Sports Med*. 2012;40(9):1985-1992.
5. Briggs KK, Steadman JR, Hay CJ, Hines SL. Lysholm score and Tegner activity level in individuals with normal knees. *Am J Sports Med*. 2009;37(5):898-901.
6. Ejerhed L, Kartus J, Sernert N, Köhler K, Karlsson J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction? A prospective randomized study with a two-year follow-up. *Am J Sports Med*. 2003;31(1):19-25.
7. Eriksson K, Anderberg P, Hamberg P, et al. A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Br*. 2001;83(3):348-354.
8. Fairbank TJ. Knee joint changes after meniscectomy. *J Bone Joint Surg Br*. 1948;30(4):664-670.
9. Forster MC, Forster IW. Patellar tendon or four-strand hamstring? A systematic review of autografts for anterior cruciate ligament reconstruction. *Knee*. 2005;12(3):225-230.
10. Gifstad T, Foss OA, Engebretsen L, et al. Lower risk of revision with patellar tendon autografts compared with hamstring autografts: a registry study based on 45,998 primary ACL reconstructions in Scandinavia. *Am J Sports Med*. 2014;42(10):2319-2328.
11. Gifstad T, Sole A, Strand T, Uppheim G, Grontvedt T, Drogset JO. Long-term follow-up of patellar tendon grafts or hamstring tendon grafts in endoscopic ACL reconstructions. *Knee Surg Sports Traumatol Arthrosc*. 2013;21(3):576-583.
12. Herrington L, Wrapson C, Matthews M, Matthews H. Anterior cruciate ligament reconstruction, hamstring versus bone-patella tendon-bone grafts: a systematic literature review of outcome from surgery. *Knee*. 2005;12(1):41-50.
13. Kellgren JH, Lawrence JS. Radiological assessment of osteoarthritis. *Ann Rheum Dis*. 1957;16(4):494-502.
14. Laxdal G, Kartus J, Hansson L, Heidvall M, Ejerhed L, Karlsson J. A prospective randomized comparison of bone-patellar tendon-bone and hamstring grafts for anterior cruciate ligament reconstruction. *Arthroscopy*. 2005;21(1):34-42.
15. Leys T, Salmon L, Waller A, Linklater J, Pinczewski L. Clinical results and risk factors for reinjury 15 years after anterior cruciate ligament reconstruction: a prospective study of hamstring and patellar tendon grafts. *Am J Sports Med*. 2012;40(3):595-605.
16. Li S, Chen Y, Lin Z, Cui W, Zhao J, Su W. A systematic review of randomized controlled clinical trials comparing hamstring autografts versus bone-patellar tendon-bone autografts for the reconstruction of the anterior cruciate ligament. *Arch Orthop Trauma Surg*. 2012;132(9):1287-1297.
17. Liden M, Sernert N, Rostgard-Christensen L, Kartus C, Ejerhed L. Osteoarthritic changes after anterior cruciate ligament reconstruction using bone-patellar tendon-bone or hamstring tendon autografts: a retrospective, 7-year radiographic and clinical follow-up study. *Arthroscopy*. 2008;24(8):899-908.
18. Lohmander LS, Englund PM, Dahl LL, Roos EM. The long-term consequence of anterior cruciate ligament and meniscus injuries: osteoarthritis. *Am J Sports Med*. 2007;35(10):1756-1769.
19. Maletis GB, Inacio MC, Desmond JL, Funahashi TT. Reconstruction of the anterior cruciate ligament: association of graft choice with increased risk of early revision. *Bone Joint J*. 2013;95(5):623-628.
20. Maletis GB, Inacio MC, Funahashi TT. Risk factors associated with revision and contralateral anterior cruciate ligament reconstructions in the Kaiser Permanente ACL Registry. *Am J Sports Med*. 2015;43(3):641-647.
21. Mohtadi NG, Chan DS, Dainty KN, Whelan DB. Patellar tendon versus hamstring tendon autograft for anterior cruciate ligament rupture in adults. *Cochrane Database Syst Rev*. 2011;(9):CD005960.

22. Øiestad BE, Engebretsen L, Storheim K, Risberg MA. Knee osteoarthritis after anterior cruciate ligament injury: a systematic review. *Am J Sports Med.* 2009;37(7):1434-1443.
23. Øiestad BE, Holm I, Aune AK, et al. Knee function and prevalence of knee osteoarthritis after anterior cruciate ligament reconstruction: a prospective study with 10 to 15 years of follow-up. *Am J Sports Med.* 2010;38(11):2201-2210.
24. Øiestad BE, Holm I, Engebretsen L, Risberg MA. The association between radiographic knee osteoarthritis and knee symptoms, function and quality of life 10-15 years after anterior cruciate ligament reconstruction. *Br J Sports Med.* 2011;45(7):583-588.
25. Peat G, McCarney R, Croft P. Knee pain and osteoarthritis in older adults: a review of community burden and current use of primary health care. *Ann Rheum Dis.* 2001;60(2):91-97.
26. Persson A, Fjeldsgaard K, Gjertsen JE, et al. Increased risk of revision with hamstring tendon grafts compared with patellar tendon grafts after anterior cruciate ligament reconstruction: a study of 12,643 patients from the Norwegian Cruciate Ligament Registry, 2004-2012. *Am J Sports Med.* 2014;42(2):285-291.
27. Pinczewski LA, Lyman J, Salmon LJ, Russell VJ, Roe J, Linklater J. A 10-year comparison of anterior cruciate ligament reconstructions with hamstring tendon and patellar tendon autograft: a controlled, prospective trial. *Am J Sports Med.* 2007;35(4):564-574.
28. Rahr-Wagner L, Thillemann TM, Pedersen AB, Lind M. Comparison of hamstring tendon and patellar tendon grafts in anterior cruciate ligament reconstruction in a nationwide population-based cohort study: results from the Danish registry of knee ligament reconstruction. *Am J Sports Med.* 2014;42(2):278-284.
29. Reinhardt KR, Hetsroni I, Marx RG. Graft selection for anterior cruciate ligament reconstruction: a level I systematic review comparing failure rates and functional outcomes. *Orthop Clin North Am.* 2010;41(2):249-262.
30. Roe J, Pinczewski LA, Russell VJ, Salmon LJ, Kawamata T, Chew M. A 7-year follow-up of patellar tendon and hamstring tendon grafts for arthroscopic anterior cruciate ligament reconstruction: differences and similarities. *Am J Sports Med.* 2005;33(9):1337-1345.
31. Roos H, Adalberth T, Dahlberg L, Lohmander LS. Osteoarthritis of the knee after injury to the anterior cruciate ligament or meniscus: the influence of time and age. *Osteoarthritis Cartilage.* 1995;3(4):261-267.
32. Sajovic M, Strahovnik A, Dernovsek MZ, Skaza K. Quality of life and clinical outcome comparison of semitendinosus and gracilis tendon versus patellar tendon autografts for anterior cruciate ligament reconstruction: an 11-year follow-up of a randomized controlled trial. *Am J Sports Med.* 2011;39(10):2161-2169.
33. Samuelsson K, Andersson D, Ahlén M, Fu FH, Musahl V, Karlsson J. Trends in surgeon preferences on anterior cruciate ligament reconstructive techniques. *Clin Sports Med.* 2013;32(1):111-126.
34. Samuelsson K, Andersson D, Karlsson J. Treatment of anterior cruciate ligament injuries with special reference to graft type and surgical technique: an assessment of randomized controlled trials. *Arthroscopy.* 2009;25(10):1139-1174.
35. Spindler KP, Kuhn JE, Freedman KB, Matthews CE, Dittus RS, Harrell FE Jr. Anterior cruciate ligament reconstruction autograft choice: bone-tendon-bone versus hamstring. Does it really matter? A systematic review. *Am J Sports Med.* 2004;32(8):1986-1995.
36. Thompson S, Salmon L, Waller A, Linklater J, Roe J, Pinczewski L. Twenty-year outcomes of a longitudinal prospective evaluation of isolated endoscopic anterior cruciate ligament reconstruction with patellar tendon autografts. *Am J Sports Med.* 2015;43(9):2164-2174.
37. Zaffagnini S, Marcacci M, Lo Presti M, Giordano G, Iacono F, Neri MP. Prospective and randomized evaluation of ACL reconstruction with three techniques: a clinical and radiographic evaluation at 5 years follow-up. *Knee Surg Sports Traumatol Arthrosc.* 2006;14(11):1060-1069.
38. Zhang W, Moskowitz RW, Nuki G, et al. OARS recommendations for the management of hip and knee osteoarthritis, part I: critical appraisal of existing treatment guidelines and systematic review of current research evidence. *Osteoarthritis Cartilage.* 2007;15(9):981-1000.

For reprints and permission queries, please visit SAGE's Web site at <http://www.sagepub.com/journalsPermissions.nav>.